

Intelligent Document Extraction, Validation & API Platform

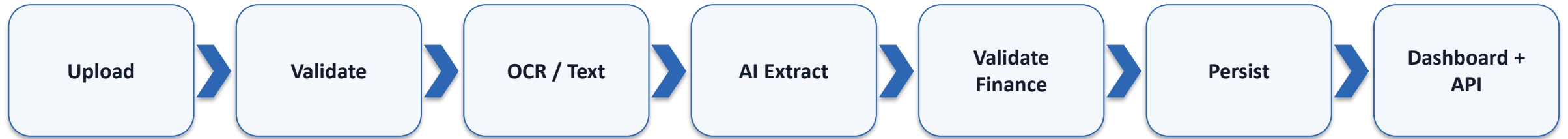
AI Engineer Internship — Technical Case Study

End-to-end financial document intelligence

Problem & Objective

- Financial teams receive invoices and financial statements in native or scanned PDF / JPG / PNG formats.
- Different layouts, image quality and OCR errors make manual extraction slow and inconsistent.
- The platform must extract complete document information, validate key financial relationships, persist results, and expose them through a dashboard and REST APIs.
- Deployment is part of the submission: both frontend and backend must be accessible through live URLs.

Solution Overview



Core design principle

LLM extracts structured values; deterministic Python rules validate financial relationships.

Documents in Scope

Invoice

Visible invoice fields + line items/table values.
Examples: number, date, parties, currency, subtotal, tax, discount, total.

Balance Sheet

Statement headers, periods, currency and every visible financial line item/value.

Profit & Loss

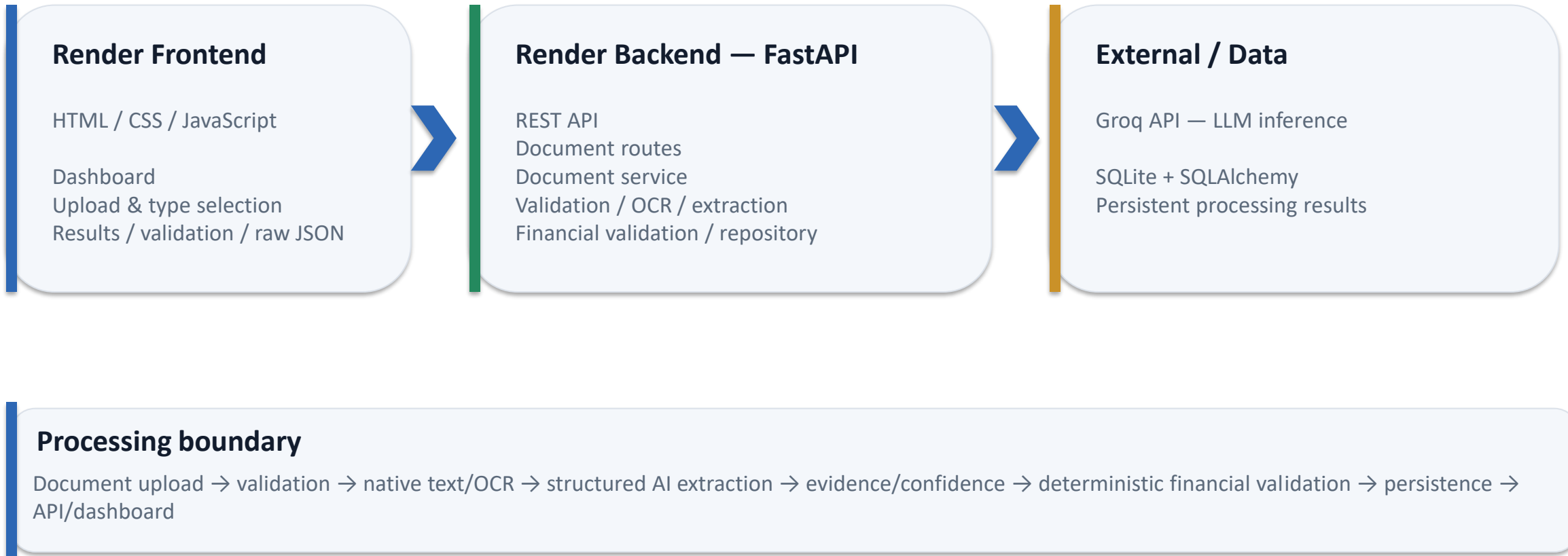
Income/expense line items and key totals such as revenue, COGS, gross profit, operating profit, tax and net profit.

Cash Flow Statement

Cash-flow line items plus operating, investing, financing, opening cash, net change and closing cash.

System Architecture

Frontend and backend deployed as separate Render services



OCR & Text Extraction

Native PDF

Extract embedded PDF text first when available.
Preserves an efficient path for text-based documents.

OCR Fallback

Scanned PDFs and image documents use Tesseract OCR with rendered/preprocessed pages.

Structured OCR Data

Text plus page metadata, words/lines, confidence and layout information support downstream grounding.

Supported inputs

PDF • JPG • JPEG • PNG • maximum 3 pages • corrupt/empty/unsupported files rejected before extraction

AI-Based Structured Extraction

Extraction Service

- Groq LLM integration
- Provider-neutral extraction boundary
- Type-aware structured Pydantic models
- Flexible fields for complete visible information
- Missing values represented as NOT_APPLICABLE / null-compatible structured values

Evidence & Grounding

Each extracted field can retain:

- source text
- page number
- bounding box where available
- confidence score

This makes important values traceable to the source.

No Silent Hallucination

The extraction contract is designed around source-supported values.

Unreadable or unavailable inputs do not become invented financial values.

Deterministic Financial Validation

Invoice

Quantity × Unit Price ≈ Line Total

Subtotal + Tax – Discount ≈ Total

Cash Paid – Total ≈ Change

Balance Sheet

Capital & Liabilities ≈ Assets

Component totals reconcile independently for each period.

Profit & Loss

Income and expenditure reconciliations

Profit-before-minority and attributable-profit checks

Cash Flow

Operating + Investing + Financing + FX ≈ Net Increase

Opening + Net Increase + Adjustments ≈ Closing

Validation output

Rule • Formula • Inputs • Calculated value • Reported value • Variance • Period • PASS / FAIL / NOT_APPLICABLE

REST API & Persistence

GET

/api/v1/health

Deployment health check

POST

/api/v1/documents/process

Upload + process document

GET

/api/v1/documents

List processed documents

GET

/api/v1/documents/{document_name}

Retrieve latest stored result

Persistence

SQLAlchemy ORM

Documents store:

- file validation
- OCR/text extraction
- structured extraction
- financial validation
- timestamps/status

Supports dashboard history and retrieval.

Frontend Dashboard

Upload & Process

Document type selection
Cash Flow/Profit Loss / Balance
Sheet/Invoices
upload
Loading and controlled error states

Dashboard

Persisted document history
Document name / type / status
Open result pages

Result View

Structured fields
Evidence & confidence
Financial validation
Collapsed raw JSON

Security-conscious rendering

DOM APIs + `textContent` are used for API-controlled values; no innerHTML-based rendering.

Testing & Verification

160

tests passed

0 failed • 0 errors

Automated Coverage

File validation
Financial PASS / FAIL / N/A
OCR and extraction contracts
Groq structured extraction mocks
API flows
Database round-trip
Duplicate / rollback / controlled errors

Real Dataset Verification

Invoice JPG — OCR PASS
Balance Sheet PDF — OCR PASS
Profit & Loss PDF — OCR PASS
Cash Flow PDF — OCR PASS

Scanned/image-based handling verified.

Real Groq document processing was intentionally not run without explicit approval to transmit document text externally.

Deployment & Submission

- Frontend and backend deployed as separate Google Cloud services.
- Backend exposes FastAPI REST endpoints and Swagger/OpenAPI.
- Public GitHub repository contains the complete source code.
- README documents setup, architecture, API, persistence, validation, deployment and limitations.
- Final live URLs and repository URL are added after deployment verification.